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NDIC HTTP Protocol for Sending All Changes on Occurrence

Release 1.1rev1

Národní Dopravní Informační Centrum

2023-03-09

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Identifier and version:

uri

cz-ndic_push-v1.1

Protocol

1.1

Specification

1.1rev1

Updated

2023-03-09

Publisher and author:

Publisher

Národní Dopravní Informační Centrum

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This document specifies the so-called PUSH protocol. The PUSH protocol enables the sending of messages containing information from the provider's system to the recipient's web service at the moment when they are updated. This protocol is used either to send complete data snapshots when the entire snapshot is updated or to send updates of individual records."

TERMS AND ABBREVIATIONS

provider

an entity that provides messages to subscribers

subscriber

an entity that receives messages from the provider

message

a complete data object containing one or more records, transmitted between the provider and the subscriber in a single step.

record

comprehensive information of a certain type with its own identifier and a time validity.

change record

a record that provides an updated description of a record with the same identifier that the subscriber previously received.

change message

a message that contains several new or changed records, thus not conveying a complete view of the state of all records.

snapshot

a message that describes a comprehensive view of all relevant records at a given moment.

curl

cURL is a command-line tool that allows, among other things, the initiation of HTTP communication with a web server. Users can send HTTP requests, modify headers and data, and receive responses.

source

a service that enables the provision of messages by a specific provider, for a specific region, in a specific message format, and using a specific protocol.

subscription registry

a web application available at <https://mobilitydata.rsd.cz>, allowing subscribers to create accounts and manage message subscriptions from various sources. The subscription registry also enables subscription management on the provider's side (subscription approvals, etc.).

subscription

a record in the subscription registry describing the parameters of what data the subscriber wants to receive from a source and how it should be done.

acceptable duration of (subscriber's system) unavailability

the time during which the subscriber's system can be unavailable, while the provider's system continues to attempt to reestablish previous communication. Exceeding this time will block the subscription, and the provider will cease to send data. This time is usually 24 hours but may vary.

maximal interval for repeated delivery attempts

the longest interval between repeated attempts to resend a message to an unavailable subscriber's system. This time is typically 15 minutes but may vary.

ARCHITECTURE

This chapter describes the key systems and actors involved in providing information using the PUSH protocol.

2.1 External Actors

External actors do influence or utilize the described protocol in some way but are not the main focus of this specification.

An external information source directly or indirectly updates records in the provider's database.

The subscriber's application utilizes the information received from the provider in some manner on the subscriber's side.

2.2 Provider

The provider ensures the operation of the system and the information provisioning processes.

The provider's administrator approves subscriptions in the subscription registry created by a subscriber.

The subscription registry configures the delivery process.

The database with records stores current information and notifies the delivery process of changes.

The delivery process ensures the delivery of messages with records to subscriber's URLs for approved subscriptions.

2.3 Subscriber

The subscriber requests the delivery of information (by creating a subscription) and manages the systems and processes for their reception.

The subscriber's administrator configures the URL (subscriber's URL) in the subscription registry for data reception.

The subscriber's URL (or handling system) receives messages with change records, stores them in the record database, and confirms their reception to the provider.

The database with records (for subscribers) stores received records to provide awareness to its applications of currently valid records.

MESSAGE, RECORDS, AND THEIR VALIDITY

A record is comprehensive information of a certain type with its own identifier and a time validity, such as information about a road closure at a specific location, valid for a certain period. Because the description of reality can change over time, a record can be updated. This modified record will have the same identifier but an altered description of the reality.

Records are always transmitted as part of a message, which can contain one or more records in their currently valid state.

3.1 Interpretation of Records in a Message

This protocol assumes the use of a specific data format within a single subscription, but this format is determined at the data subscription level and is not defined by the protocol itself. The interpretation of what constitutes a record in the sent message and how long it remains valid needs to be inferred from the specification of the used message format.

3.2 Records Without Specified Expiration

Some records, such as expected travel times at a particular section, only mention the moment when the fact was recorded and do not specify an end to their validity. In such cases, the sender and receiver must make certain assumptions about the record's validity, for example, 10 minutes from the declared moment of recording. The assumed validity duration may not necessarily match between the provider and the subscriber. The provider must ensure that the assumed validity allows them to send messages to subscribers under normal conditions. Subscribers determine their own assumed validity based on the nature of the provided information and its usage.

3.3 Records with Explicitly Specified Long Validity

Certain information, such as road closures, may explicitly state their validity, which can sometimes be quite long (even several years).

3.4 Natural Expiration of Record Validity

An individual record will cease to be valid at the moment indicated (or assumed) in the last version of the record as its validity end.

3.5 Premature Expiration of Record Validity

However, if a record's validity, which has been currently valid, is shortened into the past, it is necessary to provide the subscriber with this new record so that they can also update the validity of the mentioned record on their side. Failure to send or process a record with a shortened validity results in the subscriber retaining records as valid, even though they are no longer valid.

3.6 Expiration of Records with Short Validity

If all records have zero or relatively short validity (e.g., information about estimated travel times at a particular section), there is no need to address premature expiration of validity since records on the subscriber's side become invalid in a very short time.

3.7 Expiration of Records with Long Validity

When a record may have a relatively long validity over time, such as road closures, scenarios of premature expiration of validity need to be addressed thoroughly, particularly with consideration of the scenario *Premature Expiration of Record Validity*.

TYPICAL PROCESS OF ESTABLISHING AND OPERATING A SUBSCRIPTION WITH THE PUSH PROTOCOL

4.1 Implementation of the Subscriber System

The subscriber implements a web service capable of receiving HTTP POST requests at the subscriber's URL (specific to a particular subscription). The request body contains the provided message, which the subscriber stores for further processing.

This web service is then exposed at a publicly accessible address. It is desirable for the domain to be secured using TLS (thus accessible via the HTTPS protocol) and to implement authentication through Basic authentication (per RFC 7617). The subscriber should verify, for example, using tools like curl, that the service is accessible and functional. For further purposes, they should make a note of the target URL and the username and password for authentication.

4.2 Subscription Establishment

The subscriber visits the Subscription Registry, creates an account if necessary, and proceeds to create a new subscription. For the selected source, which offers the desired content and uses the PUSH protocol, they primarily enter the URL to which the provider will send data and provide a username and password. These details correspond to the URL and username and password that the subscriber's system is ready to serve.

4.3 Activation of Delivery

Upon activating the established subscription (which may take some time in the case of internal approval by the provider), the provider sends the first batch of messages containing records that are valid at that moment.

4.4 Ongoing Delivery of Changes

From this point onwards, the provider's system awaits an internal update of the provided content, and as soon as this occurs, it sends the subscriber a new message.

UNAVAILABILITY AND OTHER SPECIAL SCENARIOS

5.1 Short-Term Unavailability of the Subscriber System

If, for any reason, the subscriber system becomes temporarily unavailable (connectivity outage, system restarts, brief downtime, etc.), the provider's system attempts to resend messages repeatedly, with the interval between retries gradually increasing. Once the subscriber system's availability is restored, temporarily undelivered messages are then delivered. However, records that naturally expired during the unavailability period may not be sent.

5.2 Subscriber System Unavailability Exceeding the Tolerated Duration

If the subscriber system's unavailability duration surpasses the tolerated unavailability period, the delivery is completely suspended.

5.3 Manual Suspension of Subscription

The Subscription Registry allows subscribers to manually suspend their subscription.

5.4 Resuming a Suspended Subscription

A subscriber can resume a suspended subscription in the Subscription Registry. Messages containing records that were created or changed during the suspension period and are still valid are sent at the beginning. Records that were created or changed during the suspension period but are no longer valid upon resumption are not sent. The subscription then continues in the normal mode.

5.5 Resetting Subscriptions

A subscriber can perform a subscription reset through the Subscription Registry. This immediately results in an initial batch of messages containing all currently valid records, similar to the messages sent during activation. Subsequent messages continue to be sent when records change.

REQUIREMENTS FOR THE SUBSCRIBER'S URL

This chapter describes the requirements that a service handling the subscriber's URL, where messages are sent by the provider, must meet.

6.1 HTTP POST and Status Code 200

The URL must accept HTTP POST requests, with the request body containing the sent message.

The received message must be responded to with an HTTP status code of 200 - OK.

6.2 Support for Content-Encoding: gzip

If the subscription was configured with content delivery using gzip compression, the service must accept content declared with the "Content-Encoding: gzip" header.

6.3 Authentication, Authorization, and HTTPS

The PUSH protocol does not strictly require the use of HTTPS and Basic Authentication. However, if these are not implemented, the subscriber's system is at risk of receiving unauthorized content.

The URL is advised to run over HTTPS.

The URL is advised to be secured using Basic Authentication as prescribed by RFC 7617.

QUESTIONS AND ANSWERS

7.1 How do I know if the downtime has exceeded the tolerated period?

Indicative (non-binding) parameters:

- Acceptable duration of (subscriber's system) unavailability: 24 hours
- Maximal interval for repeated delivery attempts: 15 minutes

If, after the recovery of the subscriber's system, no message is delivered within 15 minutes, the tolerated downtime has likely been exceeded.

7.2 The subscriber's system was briefly unavailable, how should I proceed?

After the recovery of the subscriber's system, the subscriber should take the following steps.

Wait for approximately 15 minutes. If at least one message is received within this time, it was a tolerated downtime, and everything should work without further action.

If messages are not coming in, there has been a permanent blockage of the subscription. It is necessary to use the subscription recovery feature in the subscription registry (this is not the creation of a new subscription, just a reactivation).

If all the received information has a relatively short validity period (e.g., 10-20 minutes), the risk of expired messages on the subscriber's side or missing messages is practically zero, and no action is required.

However, if the information has a longer-term validity, a so-called subscription reset is necessary. For more information, see 'What is a subscription reset and how to use it?' _.

7.3 How to handle a situation where the subscriber's database contains records that should no longer exist?

If there is suspicion that there are records on the subscriber's side that appear to be outdated (their validity should have been reduced, but this did not happen), there may be two reasons:

- The subscriber failed to process some received records correctly (e.g., a program error).
- The subscriber's system was temporarily unavailable with a subsequent suspension of delivery. During the suspension, the validity of some records was reduced, but the subsequent resumption of delivery did not include these changes (naturally).

In this case, a subscription reset is required. For more information, see 'What is a subscription reset and how to use it?' _.

7.4 How to address a situation where the subscriber's database is missing records that should be there?

If there is suspicion that records are missing on the subscriber's side, the cause is usually an error in processing received records on the subscriber's side.

Note: Resuming a subscription after it has been suspended sends only valid messages. Therefore, the resumption of a suspended subscription should not be the cause of this problem.

In this case, a subscription reset is required. For more information, see 'What is a subscription reset and how to use it?'

7.5 What is a subscription reset and how to use it?

A subscription reset is an event that terminates the sending of previously sent messages and reinitializes the entire sending process, so messages are sent again as if a new subscription had been created (and approved).

In practice, this means that after the reset, messages with all currently valid records are immediately sent, and subsequently, change messages are sent when there are modifications or new records.

The reset can be initiated for an active subscription in the subscription registry.

It is desirable that the subscriber, as part of the reset, removes from their database records that were received earlier and now appear to be currently valid. This is particularly important if there is a suspicion that long-valid records are persisting in the subscriber's database and should probably have already been terminated.

BIBLIOGRAPHY

RFC 7617

The 'Basic' HTTP Authentication Scheme. <https://datatracker.ietf.org/doc/html/rfc7617>